Network-Aware Testing Allocation for Infectious Disease Control Under Limited Resources:

The Role of Community Structure and Incomplete Contact Information

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Abstract

We study testing allocation on contact networks with unobserved infections and incomplete contacts. Controlled 50-day static stochastic block-model (SBM) experiments vary community structure and missing edges. To study temporal information without truncating epidemics after only three recorded days, we extend the measured Haslemere sequences through a *stochastic day-block contact model*: days 1–3 use their original ordered five-minute records, while each subsequent day independently resamples one complete observed daytime sequence. Infection transmission and Bayesian filtering operate at five-minute resolution over 50 simulated days, and testing uses only past contacts. A simulation-trained logistic infection-risk model informs two budget-constrained expected-prevention allocation rules, compared with the original adaptive heuristic and conventional testing baselines. The extension preserves the four-state susceptible–hidden infectious–confirmed–recovered (SHCR) model, without adding a latent exposed state. Observed physical contacts, assumed resampled future contacts, and all simulated infections are distinguished explicitly. Results are conditional on an assumed resampling mechanism, not forecasts of Haslemere's true later contacts or COVID-19 outcomes.

1 Introduction

In the absence of vaccines and antiviral medication, which are often unavailable at the onset of an epidemic, non-pharmaceutical interventions (NPIs) implemented in response to emerging respiratory viruses are the only option available to delay and moderate the spread of the virus on a population, especially one as interconnected as today's communities. NPI measures include case isolation, tracing and quarantining of contacts, use of personal protective equipment and hygiene measures, and policies designed to encourage physical distancing. When confronted with an epidemic, governments must make decisions under rapidly changing situations with limited information, trying to identify not only infected individuals but also the contacts through which infections may spread. While testing can identify hidden infections and prompt isolation, repeated population-wide screening is often infeasible. Because of the high costs of testing, there is a clear need for sustainable strategies that limit disease transmission.

Social contact networks provide a natural representation of this test allocation problem under a limited testing budget. However, a static network summarizing the connections may discard the ordering and frequency of encounters, while a dynamic network may be incomplete or delayed. Both sources of information loss affect the decision-making process: temporal contact patterns determine which transmission opportunities arise before or after a testing decision, and missing edges or vertices may connect otherwise separated groups. Hence, the central decision is whom

to test and when: screening an infectious person before they encounter susceptible contacts may prevent further infections, whereas screening the same person after those contacts have occurred may have little immediate effect.

In general, we know that testing contributes to the control of disease transmission. However, it remains unclear how real world social networks influence the effectiveness and efficiency of such approaches. This raises the question: **Under limited testing capacity, how do network community structure, incomplete contact information, and temporal contact history affect the performance of network-aware testing allocation relative to simpler screening policies?**

We study this question using a stochastic susceptible–hidden infectious–confirmed–recovered (SHCR) epidemic model, inspired by Huang, Ke, and Jin [1]. Two complementary network settings are used to separate our analysis of community structure and incomplete contact information with temporal contact history. First, in controlled 50-day stochastic block-model (SBM) simulations, we vary community separation and the fraction of edges visible to the testing policy. Second, in an empirical-network experiment, we use three days of Haslemere five-minute proximity measurements [2] and simulate a 50-day dynamic contact process. Disease transmission uses the true simulated network, while allocation depends only on contact history, diagnoses, and test outcomes available at each decision time. Overall, our work makes the following contributions:

1. We quantify how community separation and missing observed edges affect testing-policy performance, with paired epidemic simulations that distinguish absolute infection reduction from the incremental advantage over an adaptive heuristic.

2. We combine approximate Bayesian filtering with an individual-level infection-risk model fitted on separate simulated trajectories, and compare budget-constrained testing optimization against random, degree-based, recent-contact, and existing adaptive-heuristic screening. We explicitly evaluate predictive accuracy separately from downstream control performance.

3. We preserve Haslemere's within-day temporal ordering in a longer-horizon simulation and compare testing based on recency-weighted versus cumulative past-only contact histories.

2 Related Work

Network epidemics and localized interventions. Contact-network epidemic models represent transmission opportunities at the individual or group level. Firth et al. [3] use the Haslemere proximity network to compare the tradeoffs between epidemic control and quarantine burden for interventions, such as isolation and contact tracing. They choose to aggregate contacts into a weighted static network, while we preserve the ordering of measured five-minute contacts to create a temporal network. Kretzschmar et al. [4] further show why delays are important to the effectiveness of contact-tracing. These studies motivate timely network-based interventions, but our primary comparison concerns budget-constrained selection of individuals for testing, rather than prespecified tracing and quarantine coverage.

Testing allocation and unobserved infection states. Huang, Ke, and Jin [1] formulate a testing-budget allocation problem on a county commute network using a deterministic SHCR model,

epidemic-parameter estimation, and an optimization surrogate. Our setting incorporates the SCHR model but focuses on selecting discrete individuals on stochastic contact networks and evaluates how network observation affects those selections. Because infected individuals may remain undiagnosed, allocation additionally requires hidden-state inference. Altarelli et al. [5] develop Bayesian epidemic inference using belief propagation. In our paper, we use a simpler factorized filtering approximation, then fit a separate simulation-trained risk model. Rather than optimizing for a global multi-day policy, our optimization selects the best eligible tests for an additive one-step surrogate, and we compare against an adaptive testing heuristic as our baseline.

Temporal networks and incomplete observations. Contact timing can restrict the paths along which an infection can propagate [6], and temporal and aggregated representations may produce different spreading behavior [7]. This motivates our comparison of varying temporal networks, specifically recency-weighted and cumulative contact histories under the same simulated future-day process. In practice, networks are also often incomplete as not all contacts may be recorded. Work on incomplete contact-tracing information emphasizes that such gaps can change epidemic dynamics and policy assessments [8]. Hence, we study the effect of independently omitted observed dyads in our network by allowing disease transmission on the complete network while testing policies observe a thinned network. Our contribution lies in assessing the interaction of network structure, temporal information, hidden-state estimation, and resource-constrained testing under common simulation experiments.

3 Data

3.1 Haslemere network

On March 22, 2018, the television channel BBC Four released *"Contagion! The BBC Four Pandemic"*, a documentary on influenza epidemiology to mark the centenary of the 1918 Spanish Influenza pandemic [2]. A network capturing the interpersonal distances between 469 volunteers from Haslemere, England, collected through smartphone-based location information over three consecutive days (excluding the hours between 11pm and 7am), was used to simulate the "Haslemere outbreak" featured in the television program. The Haslemere dataset consists of 102,831 pairwise distance records indexed by participant pair and five-minute observation slot. We define a potential transmission contact when a recorded pair is at most 4 meters apart. This threshold yields 17,363 qualifying proximity records across 1,262 unique pairs. The three recorded days are the only measured contact sequences used here; future contacts and all disease outcomes are simulated, as described further in Section 4.1.

Although the Haslemere dataset is among the most comprehensive of its kind, it is still subject to a number of limitations. The data describes proximity, not diagnosed infections or documented transmission events. The volunteers only compose approximately 4.2% of the town's population, where contacts with nonparticipants are thus absent [2, 3]. Additionally, the interaction patterns are biased by participant age, since children under 13 were not allowed to participate and the rate of smartphone ownership declines substantially in adults over 65. The observed contact network should therefore not be interpreted as the complete true contact structure of Haslemere.

3.2 Controlled synthetic network setting

To separate the role of community structure from idiosyncrasies of one empirical network, we also generate synthetic undirected networks using stochastic block models (SBMs). Each network has 180 individuals allocated equally to three communities. We vary within- and between-community connection probabilities while holding expected degree at eight. Unlike Haslemere, these SBM edges are not observed data; instead, they are generated under specified network parameters. This complementary design permits controlled comparisons across community strengths and levels of missing observed edges.

4 Methodology

4.1 Static and temporal network construction with missingness

To investigate how community structure affects testing allocation, we generate synthetic social networks using a stochastic block model (SBM). We consider a population of $n = 180$ individuals, divided into three equally sized communities of 60 individuals each. Let $z_i \in \{1, 2, 3\}$ denote the community membership of individual i. For each pair of distinct individuals i and j, we generate an undirected edge according to

$$A_{ij} \sim \text{Bernoulli}(p_{z_i z_j}), \tag{1}$$

where $A_{ij} = 1$ indicates a connection between individuals i and j. Conditional on the community assignments, edges are generated independently, with connection probabilities determined by whether the individuals belong to the same community. We let $p_{z_i z_j} = p_{\text{in}}$ if individuals i and j belong to the same community, while $p_{z_i z_j} = p_{\text{out}}$ if they belong to different communities.

We let the ratio, $r = p_{\text{in}}/p_{\text{out}}$, measure the strength of community structure. When $r = 1$, within-community and between-community connections are equally likely, so the network has no preferential within-community connectivity. As r increases, within-community connections become increasingly likely relative to between-community connections, producing more strongly separated communities.

To isolate the effect of community structure, we hold the expected degree of each individual fixed at eight across all values of r. Since each individual has 59 potential connections within their community and 120 potential connections outside their community, the expected degree is

$$\mathbb{E}[d_i] = 59p_{\text{in}} + 120p_{\text{out}} = 8. \tag{2}$$

Combining this constraint with $p_{\text{out}} = p_{\text{in}}/r$ gives

$$p_{\text{in}} = \frac{8}{59 + 120/r}, \qquad p_{\text{out}} = \frac{p_{\text{in}}}{r}. \tag{3}$$

We vary $r \in \{1, 1.5, 2, 4, 8, 12\}$ to examine how different levels of community separation influence disease transmission and the relative effectiveness of testing strategies.

In addition to community structure, we investigate the effect of incomplete network information. The disease transmission simulator uses the true adjacency matrix A, whereas a testing policy observes $\widetilde{A}_{ij} = A_{ij}M_{ij}$, where $M_{ij} = M_{ji} \sim \text{Bernoulli}(q)$. Here, q is the probability that an existing

edge is observed. For example, $q = 1$ represents complete network observation. The edge-retention indicators are sampled independently across unordered pairs of individuals.

Unlike the synthetic SBM networks, which remain fixed throughout each epidemic simulation, we construct a dynamic time-varying network for the Haslemere dataset. For each observed day $b \in \{1, 2, 3\}$ and five-minute interval $s \in \{1, \ldots, 192\}$, we define

$$A_{ij}^{(b)}(s) = \begin{cases} 1, & d_{ij}^{(b)}(s) \leq 4 \text{ m}, \\ 0, & \text{otherwise}, \end{cases} \tag{4}$$

where $d_{ij}^{(b)}(s)$ is the recorded distance between individuals i and j. Absent qualifying observations are coded as zero.

Given that we want to evaluate testing strategies over a 50-day epidemic, we extend the three-day contact network through a stochastic daily block-resampling approach. Without discarding within-day contact ordering, we let J_d denote the selected source day:

$$J_d = \begin{cases} d, & d \leq 3, \\ \text{Uniform}\{1, 2, 3\}, & d = 4, \ldots, 50. \end{cases} \tag{5}$$

The network on simulated day d at five-minute interval s is then defined as

$$A_d(s) = A^{(J_d)}(s). \tag{6}$$

Each new simulated day independently reuses an entire measured daytime contact sequence, preserving its internal ordering. Importantly, the testing policy receives only the past edge observations, never the future sampled blocks. As in the SBM experiments, we can also introduce incomplete network observation by randomly removing a fixed subset of contact pairs from the information available to the testing policy, while retaining the complete simulated contact sequence for disease transmission.

4.2 Network-dependent SHCR disease model

Similar to Huang et al. [1], we extend the SIR disease model to four states, dividing the infected state into H and C since an infected individual may not be diagnosed: S (susceptible), H (hidden), C (confirmed), or R (removed, including the dead, recovered and vaccinated cases). An infected individual moves directly from S to H, and we model infection transmission using an independent-hazard assumption, under which each hidden-infectious neighbor contributes a constant transmission hazard during a given time interval. The probability that a susceptible individual escapes infection from all infectious neighbors is therefore the exponential of the negative total hazard, yielding the following network-dependent infection probability.

On the static SBM network, the daily probability that susceptible individual i becomes infected is

$$\Pr\{S_i \to H_i \mid X(t), A\} = 1 - \exp\left[-\tau \sum_j A_{ij} \mathbf{1}\{X_j(t) = H\}\right], \quad \tau = 0.055. \tag{7}$$

Here, A_{ij} indicates whether individuals i and j are connected, $X_j(t)$ denotes individual j's disease state at time t, and τ is the daily transmission hazard contributed by each hidden-infectious neighbor.

For Haslemere, disease transmission instead follows the network active during each five-minute interval s of day d:

$$\Pr\{S_i \to H_i \mid X(d,s), A_d(s)\} = 1 - \exp\left[-\tau_5 \sum_j A_{ij,d}(s)\mathbf{1}\{X_j(d,s) = H\}\right], \quad \tau_5 = 0.060. \quad (8)$$

Here, $A_{ij,d}(s)$ indicates contact during that interval, and τ_5 is the transmission hazard per five-minute contact. The transmission parameters $\tau = 0.055$ (SBM) and $\tau_5 = 0.060$ (Haslemere) are fixed modeling assumptions chosen to produce realistic epidemic spread in the simulations.

Hidden-infectious individuals may be diagnosed without selective testing (probability δ) or recover (probability γ). A positive screening test identifies a hidden infection with sensitivity 0.90 and moves the individual to C, preventing further transmission. For the SBM, $\delta = 0.025$ and $\gamma = 0.12$ per day; for Haslemere, $\delta = 0.14$ and $\gamma = 0.12$ per day, converted to five-minute transition probabilities (Appendix D). The spontaneous diagnosis probability δ and recovery probability γ are also fixed simulation parameters, chosen to represent different diagnosis and recovery timescales. There is no modeled nighttime transmission because nighttime contacts were not recorded in the observed Haslemere dataset, but diagnosis and recovery can occur overnight.

4.3 Approximate Bayesian filtering

Let $\mathcal{H}_t$ denote all history prior to testing at time t, including previously observed contacts, diagnoses, tests, and test results. Because individuals' true infection states are not fully observed, we maintain estimated probabilities $q_{ik}(t) \approx \Pr(X_i(t) = k \mid \mathcal{H}_t)$ for each state $k \in \{S, H, C, R\}$. We approximate these probabilities separately for each individual and update them in two steps.

First, we use observed diagnoses and test results to update our beliefs through Bayes' rule. For a previously unconfirmed individual with estimated hidden-infection probability $h = q_{iH}$, observing no routine diagnosis reduces this probability to h^-. A subsequent negative test with sensitivity s further reduces it to h^{neg}:

$$h^- = \frac{(1-\delta)h}{1-\delta h}, \qquad h^{\text{neg}} = \frac{(1-s)h^-}{1-sh^-}. \quad (9)$$

A positive diagnosis or test instead sets the probability of being confirmed (C) to one.

Second, we predict how infection probabilities evolve through the contact network. Using the updated beliefs, the probability that a susceptible individual i becomes infected is approximated by

$$\pi_i(t) = 1 - \prod_j \left\{1 - (1 - e^{-\tau})\widetilde{A}_{ij} q_{jH}(t)\right\}. \quad (10)$$

This accounts for potential transmission from each observed neighbor, weighted by that neighbor's estimated probability of being infectious. We then update the state probabilities as

$$\begin{aligned} q_{iS} &\to q_{iS}(1-\pi_i), \\ q_{iH} &\to (1-\gamma)q_{iH} + q_{iS}\pi_i, \\ q_{iC} &\to (1-\gamma)q_{iC}, \end{aligned}$$

with the remaining probability assigned to R. The temporal Haslemere model applies the same procedure at five-minute intervals using the corresponding transmission and transition probabilities. By treating individuals' hidden states as independent, the filter avoids computing the full joint posterior (Appendix A).

4.4 Fitted individual infection-risk model

Because the Bayesian filter relies on simplifying independence assumptions, we improve its estimates of the infection probabilities by fitting an L_2-regularized logistic regression model using independently simulated epidemic trajectories. For each individual at each testing decision, the model predicts the probability of being hidden infectious (H) using the Bayesian filter's estimate together with additional features: recent one-hop contacts with confirmed cases (where a hop refers to an edge between two people), two-hop exposure through the network, observed degree, time since the last test, elapsed simulation time, and estimated susceptibility. Training labels are the simulator's true hidden states; these labels are never available to the policy during evaluation.

Predictors are standardized, with the inverse regularization strength set to $C = 1$. We train separate models for the static and extended temporal experiments, using independent simulated epidemics for training and evaluation. Further details and predictive diagnostics are provided in Appendix B.

4.5 Budget-constrained testing allocation optimization problem

We prioritize testing individuals based on the expected number of immediate infections their testing could prevent. Let $\mathcal{E}_t$ denote the set of individuals eligible for testing, $p_i^H(t)$ their estimated probability of being hidden infectious (from the Bayesian filter or fitted model), and $p_j^S(t) = q_{jS}(t)$ the estimated probability that neighbor j is susceptible. For the static network, we define the expected benefit of testing individual i as

$$v_i(t) = \underbrace{p_i^H(t)s}_{\substack{\text{Probability of} \\ \text{detecting infection}}} \underbrace{(1-e^{-\tau})\sum_j \widetilde{A}_{ij} p_j^S(t)}_{\substack{\text{Expected immediate} \\ \text{infections prevented if detected}}}, \tag{11}$$

where s is test sensitivity. Given a budget of B tests, we select individuals by solving

$$\max_{\{a_i \in \{0,1\}\}} \sum_{i \in \mathcal{E}_t} a_i v_i(t) \quad \text{subject to} \quad \sum_{i \in \mathcal{E}_t} a_i \leq B, \tag{12}$$

where $a_i = 1$ indicates that individual i is selected for testing. With equal test costs and an additive objective, the optimizer selects the B eligible people with the highest scores. This optimizes immediate expected infections prevented, not total infections over 50 days, since it ignores overlapping transmission risks and downstream spread.

For the temporal Haslemere experiment, disease transmission always follows the time-varying contact network, but testing decisions relies on previously observed interactions. We compare two approaches for estimating future contact opportunities. The *recency-weighted* approach assigns greater importance to recent contacts, whereas the *cumulative-history* approach weights all past contacts equally. Both use the same fitted infection-risk model and optimization objective; only the contact-history summaries differ (Appendix D).

We also compare against simpler testing strategies, particularly an adaptive heuristic that combines recent exposure to confirmed cases, indirect exposure through contacts of contacts, and network degree (Appendix C). Specifically, the heuristic ranks individuals using

$$s_i(t) = \left[1 - \exp\left\{-\left(0.02 + 0.65 e_i(t) + 0.10 \frac{h_i(t)}{1 + d_i(t)/5}\right)\right\}\right](1 + 0.10 d_i(t)), \tag{13}$$

where $e_i(t)$ measures recent one-hop exposure to confirmed cases, $h_i(t)$ captures two-hop exposure, and $d_i(t)$ denotes observed degree. Unlike the model-based approach, this score uses manually specified coefficients and does not explicitly estimate infection probabilities or expected transmissions prevented. We additionally consider random, degree-based, and recent-contact testing baselines.

5 Experimental Design

The code and simulation materials for the experiments are available at https://github.com/annieliang111/STAT236Project.

5.1 Scenarios and parameter settings

For the static SBM experiments, we vary the community-strength ratio $r \in \{1, 1.5, 2, 4, 8, 12\}$ and edge-observation probability $q \in \{0.25, 0.50, 0.60, 0.75, 1\}$. Each 50-day simulation begins with five hidden-infectious individuals. Testing starts on Day 8, with a budget of 20 tests per day and a five-day retesting cooldown. The fitted risk model is trained on 90 independent simulated epidemics and evaluated on 24 held-out trajectories. Policy performance is assessed using 30 paired epidemic replicates per condition.

For Haslemere, we simulate 50 days of five-minute temporal contacts, using the three observed days and resampling complete daily contact sequences thereafter (Section 4.1). Each simulation begins with 25 hidden-infectious and four confirmed individuals. Testing occurs four times daily, with five tests per decision and a 192-recorded-slot retesting cooldown. The temporal risk model uses 24 training and 10 held-out trajectories. We evaluate policies on 12 paired epidemics under complete and 50% contact observation. The original three-day experiment is retained in Appendix E.

5.2 Policies and evaluation

We compare no testing, random testing, degree-based targeting, recent-confirmed-contact targeting, the adaptive heuristic, and mean-field and fitted-risk optimization. For Haslemere, we additionally compare recency-weighted and cumulative-history versions of fitted-risk optimization.

To reduce simulation variability, policies are evaluated using paired random draws for initial infections, network generation, transmission, and contact observation; Haslemere policies also share the same resampled future contact sequences. Each policy uses only information available at its testing decisions. The primary outcome is the proportion of individuals ever infected by Day 50. We report mean outcomes and paired 95% Monte Carlo confidence intervals for policy differences, reflecting simulation uncertainty under the assumed model.

6 Results

We first evaluate whether the fitted logistic regression improves hidden-infection risk prediction relative to the approximate Bayesian filter in Section 6.1. We then assess how these estimates translate into testing-policy performance, examining the effects of community structure and incomplete network observation in the synthetic SBM experiments in Section 6.2 and 6.3. Finally, we compare

testing strategies on the extended temporal Haslemere network in Section 6.4, focusing on whether recency-weighted contact information improves allocation relative to cumulative contact histories.

6.1 Held-out infection-risk prediction

Before evaluating testing policies, we assess whether the fitted logistic regression improves infection-risk estimation compared with the approximate Bayesian filter alone. On 24 independently simulated held-out epidemic trajectories, the fitted model achieves a lower Brier score (0.0455 vs. 0.0476) and log loss (0.181 vs. 0.218), indicating more accurate predicted infection probabilities. Its AUC also increases from 0.691 to 0.736, suggesting better discrimination between hidden-infectious and noninfectious individuals. These metrics are descriptive results from simulated epidemics (full table in Appendix B).

These results suggest that incorporating additional network and testing-history features improves infection-risk prediction beyond the Bayesian filter alone. However, improved predictive accuracy does not directly imply better epidemic control, since the value of testing also depends on an individual's potential future contacts. Hence, we evaluate the testing policies separately in the following sections.

6.2 Community structure and policy performance

We investigate whether stronger community structure affects the relative effectiveness of network-aware testing strategies, holding the expected degree fixed. As the community-strength ratio r increases, connections become more concentrated within communities rather than between them. Table 1 compares the mean proportion of individuals infected by Day 50 under different testing strategies at $r = 1$ and $r = 12$, representing the weakest and strongest community separation considered. The fitted-risk optimizer achieves the lowest mean infection rates compared to the other methods presented at both community-strength ratios. The mean-field-only optimizer performs worse, with infection rates of 38.1% and 36.3%, respectively.

Allocation policy	$r = 1$	$r = 12$
Random	46.7	44.8
Degree	36.5	34.1
Recent contact	37.9	31.3
Adaptive heuristic	35.0	30.4
Mean-field optimizer	38.1	36.3
Fitted-risk optimizer	32.6	30.0

Table 1: Mean percentage infected by Day 50 across 30 paired static-SBM epidemics ($B = 20$, complete network observation), shown at two contrasting community-strength ratios.

Figure 1 shows the difference in infection proportions between the fitted optimizer and adaptive heuristic across all six community-strength ratios, with negative values favoring fitted optimization. Although the point estimates suggest an advantage in the settings shown in Table 1, the paired 95% Monte Carlo intervals include zero, and there is no clear monotonic relationship between community separation and the optimizer's relative advantage. Thus, these results are insufficient to establish that stronger community separation systematically increases the advantage of fitted-risk optimization over the simpler adaptive heuristic.

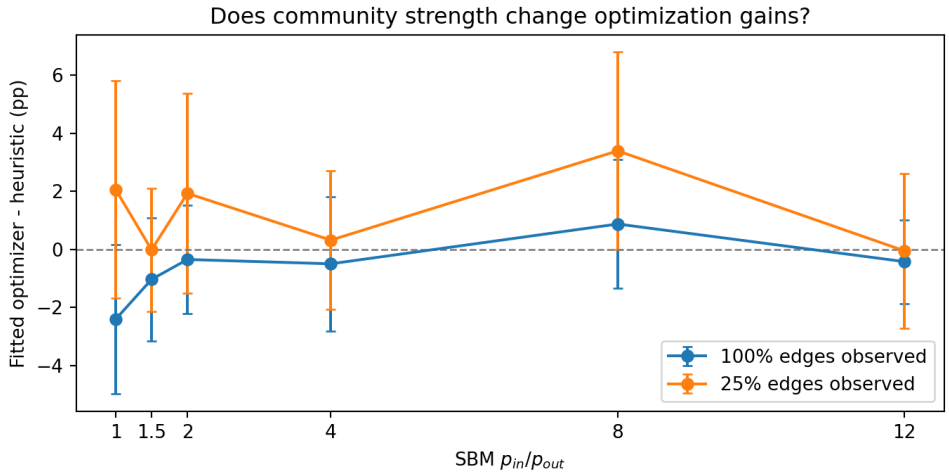


Figure 1: Fitted optimizer minus adaptive heuristic (in percentage of individuals infected by Day 50) across the six SBM community ratios; negative values favor fitted optimization. Error bars are paired 95% Monte Carlo intervals.

6.3 Incomplete observation of the network

In analyzing how incomplete contact information affects testing performance, disease transmission continues along the full network, while testing policies only observe a fraction q of existing edges. This allows us to isolate the effect of missing network information on testing decisions. As shown in Figure 2, lower edge visibility generally leads to higher infection rates under both fitted-risk optimization and the adaptive heuristic. This suggests that missing contacts reduce the effectiveness of network-informed testing, as expected.

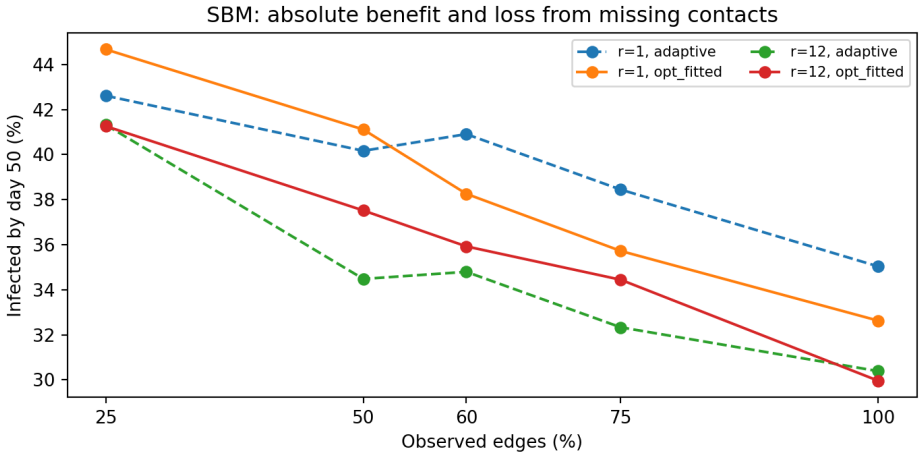


Figure 2: Simulated infection percentages under varying observed-edge fractions, contrasting fitted optimization with the adaptive heuristic on selected SBM community-strength settings. The true transmission network is unchanged by edge thinning.

However, the effect on their relative performance is less straightforward. At $r = 1$, the fitted optimizer achieves a lower infection rate than the adaptive heuristic under complete observation (32.6% vs. 35.0%). When only 25% of edges are observed, this ordering reverses, with approximately

45% infected under fitted optimization versus 43% under the heuristic. This suggests that missing contacts can affect the two strategies differently: the fitted optimizer relies on estimated infection probabilities and susceptible neighbors, whereas the heuristic uses simpler measures of exposure and connectivity. Nevertheless, the relative differences do not follow a consistent trend across observation levels or community structures. Thus, we find no clear evidence that either strategy is systematically more robust to incomplete network information.

6.4 Extended 50-day dynamic Haslemere results

Policy	Ever infected (%)	New infections	Tests used
No testing	39.98	158.5	0.0
Random	35.70	138.4	1000.0
Degree	34.84	134.4	1000.0
Recent confirmed contacts	24.70	86.8	1000.0
Adaptive heuristic	25.69	91.5	1000.0
Mean-field optimizer	28.70	105.6	1000.0
Temporal fitted optimizer	22.87	78.2	1000.0
Cumulative-history fitted optimizer	22.76	77.8	1000.0

Table 2: Mean simulated ever-infected percentage over 12 paired 50-day dynamic Haslemere epidemics with complete observed-dyad information. The 29 initially infected people are included in counts.

With complete contact information, the temporal fitted optimizer has 22.87% ever infected, compared with 22.76% for cumulative-history fitting and 25.69% for the original adaptive heuristic. The paired temporal-minus-cumulative difference is +0.11 percentage points (95% MC interval [-1.52,+1.73]), which includes zero; therefore the longer simulation still does *not* demonstrate an advantage from recency weighting. Relative to random allocation, the temporal optimizer reduces the mean simulated infection percentage by 12.83 points. With half the dyads visible, its mean infection fraction increases by 4.41 points. These Monte Carlo findings are exploratory because they use only 12 epidemic draws and are conditional on the day-block resampling model. The full network-visibility comparison is plotted in Appendix Figure 3. The fitted recency- and cumulative-history optimizers are both compared under the same dynamically changing true contact process. Their difference should be interpreted as a short-horizon allocation comparison conditional on the day-block resampling assumption, not a prediction about future observed Haslemere contacts.

7 Discussion

The controlled SBM experiments and the dynamic Haslemere experiment answer complementary questions. The former isolates community-strength and edge-visibility effects while holding the network-generating assumptions explicit. The latter asks whether a longer-run epidemic simulator that retains the ordering of five-minute contacts gains more from a recency-weighted history than from past cumulative frequency. Paired simulations hold initial infections, epidemic innovations, missing-edge masks, and the sampled future contact-block path fixed across policies.

The results do not establish that better hidden-state prediction or a more explicitly optimized surrogate always improves epidemic control relative to the handcrafted heuristic. Our one-step

objective ignores overlapping contacts and long-run cascades. Likewise, the Haslemere contact sample covers only a small, selected subset of residents; day-block resampling cannot generate new partnerships, model day-of-week shifts, or recover unobserved nighttime contacts. While diagnosis and recovery can occur overnight, no nighttime transmission is simulated. The SHCR model assumes that newly infected individuals immediately enter the infectious state, and its clinical parameters are assumed rather than estimated. Simulated missingness removes dyads at random, omitting other plausible observation biases. Consequently, these estimates describe comparisons within the chosen simulation models, not actual epidemiological effects of a testing program.

Reproduction commands, detailed filtering recursions, fitted coefficients, additional policy contrasts, and the three-day measured-only sensitivity study appear in the appendix and accompanying code.

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A Exact time ordering and approximate filtering details

Observed information. At decision time the algorithm knows $\widetilde{A}$, each previous test day and result, newly observed routine diagnoses, and previously confirmed status. It never sees the simulator's latent S, H, R states. The latent simulation carries four states for every person; routine diagnosis occurs before the screening decision, followed by instantaneous testing, disease transmission, and recovery. A person infected on day t cannot recover before day $t+1$.

For $\mathbf{q}_i = (s_i, h_i, c_i, r_i)$, a non-confirmed individual with no new routine diagnosis has prior-to-decision posterior

$$\mathbf{q}_i^- = (s_i, (1-\delta)h_i, c_i, r_i)/(1-\delta h_i).$$

A positive diagnosis forces $\mathbf{q}_i^- = (0,0,1,0)$. For a test-negative individual with current posterior $\mathbf{q}_i^-$, set

$$\mathbf{q}_i^+ = (s_i^-, (1-s)h_i^-, c_i^-, r_i^-)/(1-sh_i^-),$$

while a positive test gives $\mathbf{q}_i^+ = (0,0,1,0)$. Untested people retain $\mathbf{q}_i^+ = \mathbf{q}_i^-$. Given these post-test marginals, the predictive probability of new infection in a still-susceptible person i is π_i in (10), with $\eta = 1 - e^{-\tau}$, and

$$\begin{aligned}
s_i(t+1) &= s_i^+(1-\pi_i),\\
h_i(t+1) &= (1-\gamma)h_i^+ + s_i^+\pi_i,\\
c_i(t+1) &= (1-\gamma)c_i^+,\\
r_i(t+1) &= r_i^+ + \gamma(h_i^+ + c_i^+).
\end{aligned}$$

The marginal prediction uses the observed network and is repeated sequentially; it does not integrate jointly over correlated neighbors or learn τ, δ, γ. Initial marginal hidden probability is $5/n$, which approximates the simulator's uniform draw of exactly five seeds.

B Risk model, estimation target, and data separation

Let $Y_{it} = \mathbf{1}\{X_i(t) = H\}$ be the training simulator label after routine diagnosis and before selective testing. For eligible training person-days, define x_{it} with seven entries:

1. $\operatorname{logit}(q_{iH}(t))$ (clipped for numerical stability);
2. $\log(1 + e_i(t))$ for recent, decayed confirmed contacts;
3. $\log\{1 + h_i(t)/(1+d_i)\}$ for two-hop exposure;
4. $\log(1+d_i)$ for observed degree;
5. $t/50$, time within the epidemic;
6. $\log\{1 + \min(50, t - \ell_i)\}$, time since last test ℓ_i (capped at 50);
7. $q_{iS}(t)$, the filter's susceptible posterior.

Here $e_i(t) = \sum_j \widetilde{A}_{ij} w_j(t)$ and $h_i(t) = \sum_j \widetilde{A}_{ij} e_j(t)$, with $w_j(t) = \mathbf{1}\{0 \le t - c_j < 7\} \exp[-(t-c_j)/4]$ for recorded confirmation day c_j. The symbol $h_i(t)$ in this section denotes the two-hop feature, distinct from posterior hidden probability q_{iH}. The estimator is

$$\widehat{p}_i^H(t) = \operatorname{logit}^{-1}(\hat{\alpha} + \hat{\beta}^\top \widetilde{x}_{it}), \qquad (\hat{\alpha}, \hat{\beta}) = \arg\min \left\{ -\sum_{it} \log p_{\alpha,\beta}(Y_{it} \mid x_{it}) + \lambda \|\beta\|_2^2 \right\},$$

where predictors are standardized and the implementation uses scikit-learn logistic regression with inverse regularization strength $C = 1$ (the software's regularization convention). Random

and recent-contact testing generate the 90 independently seeded training trajectories; 24 further trajectories are reserved for held-out evaluation. Both SBM and Haslemere topologies enter the training distribution. There is no random train/test split of person-days from the same epidemic trajectory. Individual rows are still correlated within trajectories; the held-out scores are descriptive predictive metrics, not independent-sample hypothesis tests.

Prediction method	Brier ↓	Log loss ↓	AUC ↑
Mean-field filter	0.0476	0.218	0.691
Fitted logistic model	0.0455	0.181	0.736

Table 3: Held-out eligible-person prediction diagnostics (reproduced for appendix context).

Feature	Standardized coefficient
Filter log-odds of hidden infection	+0.320
Recent confirmed one-hop contacts	+0.208
Two-hop contact exposure	+0.211
Observed degree	+0.342
Day	-0.305
Time since last test	+0.207
Filtered susceptibility	+0.191

Table 4: Logistic-regression coefficients after standardization of predictors. These are estimates from simulated labeled trajectories; they are not estimates of real disease parameters.

C Testing heuristics, objective limits, and implementation

The original adaptive heuristic, retained explicitly as a baseline, ranks eligible people by

$$\left[1 - \exp\left\{-\left(0.02 + 0.65e_i(t) + 0.10\frac{h_i(t)}{1 + d_i/5}\right)\right\}\right](1 + 0.10d_i).$$

Its coefficients are design choices, not estimated epidemic parameters, and the final product is not a probability. For recent-contact screening, the score is $e_i(t)$; for degree screening it is d_i; ties are randomized by pre-generated uniforms. Random screening samples eligible nodes uniformly using the same tie-breaking innovation. The filtered and fitted optimizers both solve (12) by sorting the additive v_i, not a Frank–Wolfe program.

The marginal one-step benefit would also depend on whether a potential recipient j has other infectious neighbors and on overlapping tests; equation (12) ignores both. In addition, optimizing one-day first-generation cases prevented is not equivalent to minimizing cumulative infections, hidden person-days, or long-run welfare. Thus model-based optimizer here denotes an exact optimization of a stated approximate objective, not globally optimal epidemic control. Because the fitted estimator is trained on prior policy distributions, we evaluate the policy in new epidemic simulations rather than assessing only in-sample log loss.

Comparison	Effect (pp)	95% MC interval
$(r = 12) - (r = 1)$, $q = 1$	2.0	[-1.1, +5.1]
$(r = 12) - (r = 1)$, $q = 0.25$	-2.1	[-6.6, +2.4]
SBM $r = 12$, $q = .25$ vs 1	11.3	[+7.8, +14.8]
Haslemere, $q = .25$ vs 1	8.8	[+5.5, +12.1]

Table 5: Selected paired contrasts: community interaction in fitted-optimizer-versus-heuristic advantage, and increase in fitted-optimizer infection rates when observation drops from 100% to 25%. Positive values in the final two rows mean degraded absolute disease control.

D Extended daily-block temporal model

Network resampling. The observed 7am–11pm day comprises 192 consecutive five-minute matrices; nights span eight hours with unknown contacts. We retain Day 1, 2, and 3 as measured and sample whole source-day blocks J_d for each Day 4–50. A source-day draw is fixed across policies/visibility levels for a given paired epidemic replicate, and is not revealed to the policy before each contact arrives. We do not independently sample individual five-minute intervals because that would disrupt within-day ordering. Resampling with replacement from only three days is a limited empirical bootstrap and may understate structural variation.

Transmission and filtering. The four-state $S \to H \to C/R$ model is unchanged, with per-observed-contact hazard $\tau_5 = 0.060$. Daily baseline recovery/diagnosis probabilities are mapped to per-slot probabilities as $1 - (1 - \gamma)^{1/288}$ and $1 - (1 - \delta)^{1/288}$. The eight-hour nightly interval uses corresponding 96/288 exponents for diagnosis and recovery, without modeled transmission. An individual infected during a five-minute slot is immediately H by its end, never E. Transmission is simulated on full sampled $A_d(s)$, whereas the factorized filter predicts from only the dyad-thinned $\widetilde{A}_d(s)$ and observed diagnoses/test results. The dyad-missingness mask remains fixed throughout the 50 days.

Allocation history. Let k be the global index over 50×192 recorded slots; overnight hours are included in exponential decay but not in cumulative recorded-slot counts. For the 50-day extension we set the recency e-folding time to 12 hours, $\lambda = \exp(-1/144)$, and calculate

$$W_{ij}^{\text{recent}}(k) = \sum_{s<k} \lambda^{k-1-s} \widetilde{A}_{ij}(s), \qquad W_{ij}^{\text{cum}}(k) = \sum_{s<k} \widetilde{A}_{ij}(s),$$

with extra λ^{96} decay over each night. The next four-hour contact surrogates are $48(1-\lambda)W^{\text{recent}}$ and $48W^{\text{cum}}/\max(1,k)$, respectively. Our expected-prevention rule multiplies each candidate's fitted hidden risk, test sensitivity, per-contact transmission factor $(1 - e^{-\tau_5})$, and the susceptible-neighbor weighted surrogate. The optimizer solves the additive top-five allocation exactly each round, but does not optimize the full 50-day objective. Eligibility excludes confirmed people and anyone screened within the previous 192 recorded slots. The fitted model is newly trained on 24 independent extended trajectories using hidden-state simulation labels and held out on 10 trajectories, not reused from the three-day experiment.

Visible dyads (%)	Temporal fitted	Cumulative fitted	Heuristic
50	27.27	25.36	25.76
100	22.87	22.76	25.69

Table 6: Extended Haslemere mean ever-infected percentages under two network-observation rates, 12 paired simulated epidemics; later-day contacts are resampled.

Estimator	Brier ↓	Log loss ↓	AUC ↑
Mean-field filter	0.0239	0.0876	0.875
Fitted logistic	0.0139	0.0591	0.898

Table 7: Predictive diagnostics of the fitted hidden-infection risk model on entirely separate simulated 50-day trajectories. Person-round samples are correlated within epidemics.

Observed dyads	Comparator	Difference (pp)	95% MC interval
100%	Random	-12.83	[-16.18,-9.48]
100%	Adaptive heuristic	-2.83	[-6.07,+0.42]
100%	Cumulative optimizer	+0.11	[-1.52,+1.73]
50%	Random	-8.42	[-10.95,-5.89]
50%	Adaptive heuristic	+1.51	[-1.26,+4.28]
50%	Cumulative optimizer	+1.92	[+0.15,+3.69]

Table 8: Extended 50-day paired differences: temporal fitted optimizer minus comparator; negative indicates fewer simulated infections under the temporal rule. These intervals quantify Monte Carlo variability across 12 shared simulated epidemics, not uncertainty about real population effects.

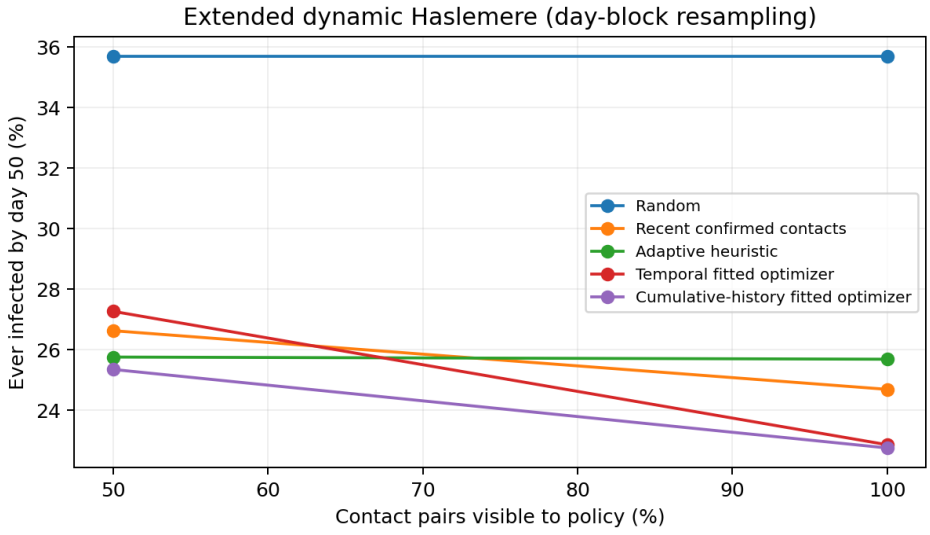


Figure 3: Infection percentages by Day 50 under complete and 50% observed-dyad information; later-day contact sequences are generated by resampling, not observed.

E Three-day measured-only sensitivity (previous experiment)

Observation and action timing. Each of the 576 Haslemere slots spans five minutes in one of three 7am–11pm recording windows. On the first slot of each four-hour testing block, routine diagnoses are observed, the filter is updated, and up to five eligible people are selected and tested. Then infections occur along current $A(k)$, old hidden/confirmed individuals may recover, the filter predicts with $\widetilde{A}(k)$, and only afterward is the slot's contact information added to the history used for future decisions. Two intervening eight-hour gaps allow diagnosis/recovery progression without modeling transmission. The same independently sampled dyad-missingness mask applies at every observed slot, preserving repeated encounters of retained dyads. Tests within the preceding 192 recorded slots are ineligible for repeat screening. In the simulator, confirmed people never transmit even though the measured contact sequence is held fixed across policies.

Temporal filtering. With $\eta_5 = 1 - \exp(-\tau_5)$, the factorized five-minute prediction computes

$$\pi_i(k) = 1 - \prod_j [1 - \eta_5 \widetilde{A}_{ij}(k) q_{jH}(k)], \quad q_{iS}(k+1) = q_{iS}(k)(1 - \pi_i(k)),$$

and updates q_{iH}, q_{iC}, q_{iR} exactly as in Appendix A with the five-minute recovery probability. Routine-diagnosis and negative-test Bayes updates also replace daily rates by five-minute rates where appropriate. The initial posterior puts hidden mass $25/(469 - 4)$ on each previously unconfirmed person and point mass on C for four already confirmed individuals. A separately trained L_2 logistic model uses filter hidden logit, one- and two-hop recent confirmed contacts, recent and cumulative contact strength, filtered susceptibility, normalized time, and last-test timing. The risk-model training has 60 epidemic trajectories; its 16-trajectory held-out performance is summarized below. Hidden-state labels never enter the test-selection input.

Method	Brier score	Log loss	AUC
Mean-field	0.1052	0.3500	0.6132
Fitted temporal risk	0.0670	0.2558	0.6179

Table 9: Independent temporal-trajectory holdout diagnostics (simulation-generated labels). Person-slot observations within a trajectory are dependent; reported scores are descriptive.

Causal recency versus cumulative histories. Let $M_{ij}(s)$ denote the edge-observation mask applied to $A_{ij}(s)$, let $\lambda = \exp(-1/24)$ and k index observed five-minute slots. The recency history before the decision at k is

$$W_{ij}^{\text{recent}}(k) = \sum_{s<k} \lambda^{k-1-s} \widetilde{A}_{ij}(s), \qquad W_{ij}^{\text{cum}}(k) = \sum_{s<k} \widetilde{A}_{ij}(s).$$

At the two intervening nights, the recency sum is additionally multiplied by λ^{96}. The cumulative history is rescaled by the number of recorded previous slots, not wall-clock time. Let p_i^H be the fitted or filtered current infection risk. The next-block value surrogate for testing i is

$$v_i(k) = p_i^H(k)\, s\, \tau_5 \sum_j \widehat{L}_{ij}(k) q_{jS}(k), \quad \widehat{L}_{ij}^{\text{recent}} = 48(1-\lambda) W_{ij}^{\text{recent}}, \quad \widehat{L}_{ij}^{\text{cum}} = 48 W_{ij}^{\text{cum}} / \max(1, k).$$

The score assumes one expected block of contacts based on the past; it is an additive heuristic objective, not a real-time forecast using future recorded contacts. Maximizing its sum under

$\sum_i a_i \leq 5$ chooses the five largest eligible scores. The fitted temporal and cumulative-history optimizers use the same current risk probability to isolate the contribution of the contact-summary rule. When $k = 0$, both matrices are zero, so tie-breaking is randomized. The comparison is not a comparison with an oracle static network built from all three days.

Observed pairs (%)	Temporal optimizer	Cumulative-history optimizer	Heuristic
25	13.94	14.71	13.96
50	13.93	13.85	14.37
75	13.94	13.48	13.70
100	13.92	13.21	13.62

Table 10: Simulated ever-infected percentage under increasing contact visibility, across 15 paired episodes. Temporal and cumulative histories are both constructed from previously observed five-minute contacts.

Comparator	Difference (pp)	95% MC interval
Random	-0.81	[-1.77, +0.15]
Recent-contact	+0.18	[-0.38, +0.75]
Adaptive heuristic	+0.30	[-0.21, +0.80]
Filter optimizer	-0.30	[-1.15, +0.55]
Cumulative-history optimizer	+0.71	[-0.11, +1.53]

Table 11: Selected paired temporal-minus-comparator contrasts in percentage points. Negative favors the fitted temporal optimizer; confidence intervals represent Monte Carlo error across 15 paired epidemics.

Sensitivity and scope. Binary within-four-meter contacts are retained per slot; repeated encounters can therefore contribute repeated hazards across slots. Transmission hazard 0.060 per observed five-minute encounter and initial disease prevalence are assumed and need empirical calibration before clinical interpretation. Contact data do not establish unrecorded contacts at night or disease outcomes. Any inference about longer-term performance, unobserved contacts, or changes in behavior would require additional assumptions and data.

F Reproduction and limitations of parameter identification

The primary extended 50-day study uses `extended_temporal_haslemere.py` with 24 training trajectories, 10 held-out trajectories and 12 paired evaluation replicates. The original measured-only three-day sensitivity uses `temporal_haslemere.py` with 60 training, 16 held-out and 15 evaluation trajectories. Both use the original public proximity CSV; the fitted risk models and output files are separate. More Monte Carlo replications and alternative block-resampling schemes should be considered before drawing strong conclusions.

1. The original public `Kissler_DataS1.csv` (469 IDs; 102,831 rows) is already included in the `data/` folder. The network loader is `run_haslemere.py`.
2. Run `fit_risk_model.py` (seed 339236) and inspect `models/risk_model.joblib`, held-out scores and coefficients.
3. Run `run_structure_sensitivity.py` and `run_haslemere_visibility.py` to recreate the four conventional-policy baseline trajectories, then `run_optimized_experiments.py` to add paired

optimized-policy outcomes (evaluation seed 2362026). The optimized runner verifies that the baseline files contain the requested replicate IDs and uses cached outputs to resume interrupted jobs.

4. Generate the tables with `create_model_based_assets.py`; compile with `pdflatex`, `biber report`, then `pdflatex` twice. The `analysis.Rmd` notebook and `index.html` are supplementary deliverables.

The fixed epidemic parameters describe a chosen data-generating process, and cannot be identified from physical proximity alone. Estimated logistic coefficients describe hidden-infection risk under that simulated process, rather than real-world clinical transmission parameters. Limitations include network missingness at random, no observation error on retained edges, no test false positives, no reporting delays, no behavioral response, and no transmission calibration. Generalizing would require observed disease trajectories, different network samples, and more realistic delays and sampling designs.